Algebraic Multiplicity Does Not Determine the Perturbation Exponent: A Disproof of Conjecture 00000005131

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Abstract

The zero matrix of order two has an eigenvalue of algebraic multiplicity two, but its worst perturbational eigenvalue displacement is exactly ε, and its pseudospectral area is exactly $\pi\varepsilon^2$. These are not the proposed sharp powers $\varepsilon^{1/2}$ and $\varepsilon^{2/2}$. Lean verifies the actual complex matrices, all admissible perturbations, the characteristic polynomial and multiplicity, the exact pseudospectrum, its Lebesgue area, and the negations of the corresponding universal power laws.

Reading of the source. Both source languages explicitly introduce m as *algebraic multiplicity*, assert displacement of $\varepsilon^{1/m}$ type, and assert pseudospectral area of $\varepsilon^{2/m}$ type. We interpret a stated scaling exponent as a sharp power: comparability up to positive constants, or a positive finite leading asymptotic constant. We refute even the lower bound necessary for either meaning. A mere big-O upper bound is a weaker statement and is not refuted here.

The same source says the exponent is decided by the maximal Jordan block size. That is the correct distinction, and it conflicts with using algebraic multiplicity as the exponent denominator. Our result does *not* refute the version in which m is consistently replaced by maximal Jordan block size. In the example that size is one. The independent, explicit area clause also fails with $m = 2$, so this is not an argument based only on one chosen perturbation path.

Precise conventions. For a complex matrix E use the induced infinity norm

$$\|E\|_\infty = \max_i \sum_j |E_{ij}|.$$

Its closed perturbational pseudospectrum is

$$\Sigma_\varepsilon(A) = \{z \in \mathbb{C} : \text{some } E \text{ satisfies } \|E\|_\infty \le \varepsilon \text{ and } (A+E)v = zv \text{ for some } v \neq 0\}.$$

Area means ordinary planar Lebesgue measure. The example has only one spectral cluster, avoiding any distinction between local and total pseudospectral area. Open perturbation balls give the same area and the same supremal displacement.

Theorem 1. *Let $A = 0 \in M_2(\mathbb{C})$ and $\varepsilon \ge 0$. Then*

$$\chi_A(X) = X^2, \qquad \Sigma_\varepsilon(A) = \{z : |z| \le \varepsilon\}, \qquad \operatorname{area}(\Sigma_\varepsilon(A)) = \pi\varepsilon^2.$$

In particular the eigenvalue zero has algebraic multiplicity two, and the maximum possible eigenvalue displacement is exactly ε.

Proof. The characteristic polynomial is $\det(XI_2) = X^2$, so the root multiplicity of zero is two. Let $Ev = zv$ with $v \neq 0$, and choose a coordinate i with $|v_i| = \max_j |v_j| > 0$. Every perturbation in the stated ball satisfies

$$|z|\,|v_i| = \left|\sum_j E_{ij} v_j\right| \leq \sum_j |E_{ij}|\,|v_j| \leq \left(\sum_j |E_{ij}|\right) |v_i| \leq \varepsilon |v_i|.$$

Thus $|z| \leq \varepsilon$. Conversely, if $|z| \leq \varepsilon$, take $E = zI_2$. Its row sums are $|z|$, and it has eigenvalue z with nonzero eigenvector $(1, 1)$. This proves the exact disk description, including attainment of displacement ε at $z = \varepsilon$. Its Lebesgue area is $\pi\varepsilon^2$. □

Theorem 2. *Neither the sharp displacement exponent $1/m$ nor the sharp pseudospectral area exponent $2/m$ is universally determined by algebraic multiplicity m.*

Proof. Use the preceding matrix, with $m = 2$. If the claimed displacement exponent were sharp, some $c, \delta > 0$ would imply $c\sqrt{\varepsilon} \leq \varepsilon$ whenever $0 < \varepsilon < \delta$. Dividing by $\sqrt{\varepsilon}$ would give $c \leq \sqrt{\varepsilon}$ at arbitrarily small positive ε, a contradiction. Similarly, the area law would require $c\varepsilon \leq \pi\varepsilon^2$ on a positive interval, which would give $c \leq \pi\varepsilon$ there, again impossible.

The Lean statements use $\varepsilon = t^m$ and $t \downarrow 0$ to express both necessary lower bounds without fractional powers. The response lower bound is ct and the area lower bound is ct^2. For the example the exact values are t^2 and πt^4, respectively. All quantifiers over positive constants and all sufficiently small parameters are retained in the proofs. □

Norm and Jordan-block scope. The same counterexample applies to the customary Euclidean operator norm: the eigenvector estimate $|z|\|v\| \leq \|E\|\|v\|$ and $\|zI_2\| = |z|$ give the identical disk. Lean specifically verifies the induced infinity norm version above; this Euclidean norm observation is a paper proof. Both are standard fixed norms, and finite-dimensional equivalence of norms cannot change these sharp exponents. The zero matrix is diagonal and has maximal Jordan block size one, exactly consistent with the corrected exponent. No claim is made about a different corrected Jordan-block formulation or the source's condition-number and lattice-configuration clauses. Falsity of the universal area clause with algebraic m already refutes the stated conjunction.

Formal verification and scope. The Lean development uses Mathlib matrices over $\mathbb{C}$, actual matrix-vector multiplication, Mathlib characteristic polynomials and polynomial root multiplicity, and Mathlib planar volume. The predicate `Budget` is the row-sum characterization of the induced infinity norm ball; `Eigen` is the ordinary nonzero eigenvector equation. The exact disk theorem quantifies over *all* matrices in that ball. The principal results are

`zero_pseudospectrum`, `sharp_response`, `zero_multiplicity`,
`zero_area`, `not_multiplicity_area_law`,
`not_multiplicity_response_law`.

The area law is universally quantified over single-cluster matrices with $\chi_A = (X - a)^n$, so algebraic multiplicity is n and no unrelated eigenvalue can contribute extra area. It states only the positive lower bound required by the source's area scaling claim; negating that necessary consequence suffices. The response law similarly includes actual characteristic-polynomial multiplicity. Only the standard axioms `propext`, `Classical.choice` and `Quot.sound` occur. Development and self-review were performed by a delegated AI agent; the coordinating agent reviews the result separately. No external independent review is claimed.

Source and reproduction. The preserved English and Chinese source is `SOURCE.md`, from the repository's conjecture 00000005131. The provenance file records the upstream commit and source hash. The Lean project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the `lean` directory run

```
lake build
lake env lean -DwarningAsError=true Main.lean.
```

The source ends with axiom reports for the principal theorems.